Activity - 1

12/10/2020

① Three process arrive at time 0:

process	Burst time
P_1	5
P_2	3
P_3	8

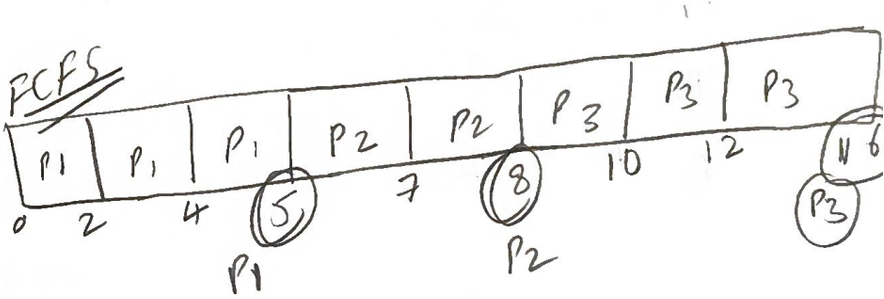
using FCFS, find the average waiting time.

Given

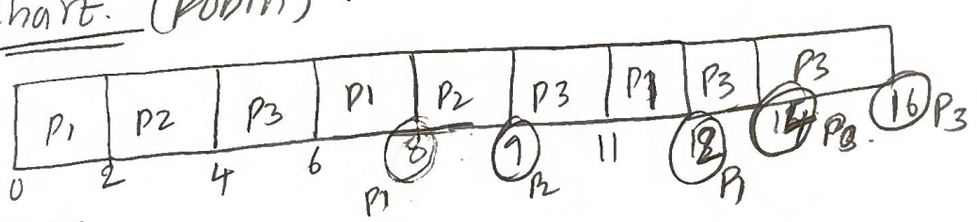
process	Burst time
P_1	5
P_2	3
P_3	8

and time quantum = 2

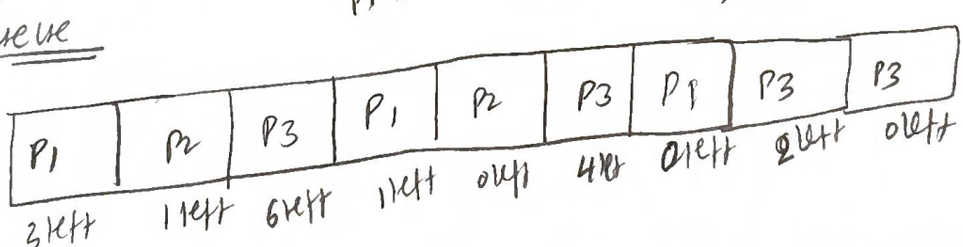
FCFS



Gantt chart. (Robin)



Ready queue



completion − arrival turn-burst

∴

process	Burst time	arrival time	Completion time	Turnat	waiting time
P_1	5	0	12 (5)	12 (5)	12−5 = 7 (0)
P_2	3	0	9 (8)	9 (8)	9−3 = 6 (5)
P_3	8	0	16 (16)	16 (16)	16−8 = 8 (8)

∴ average waiting time = $\frac{7+6+8}{3} = \frac{21}{3} = 7$ units

$= \frac{8+5}{3} = \frac{13}{3}$

average waiting time $= 3)13(4.33$

$\frac{12}{10}$

$= 4.33$

(2) Consider Round Robin with time quantum = 2

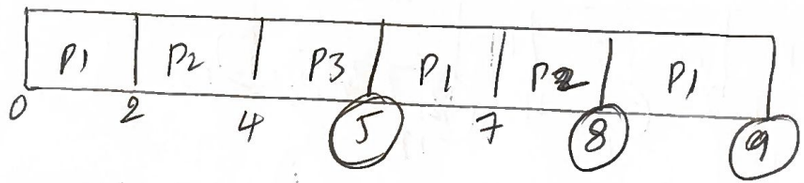
Process	Burst
P_1	5
P_2	3
P_3	1

All arrive at time 0. which process completes first

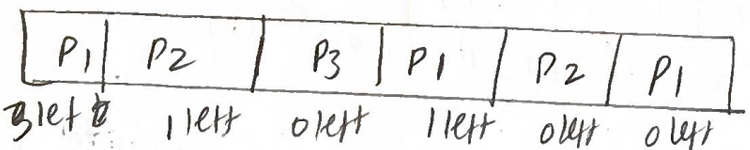
Given.

process	Burst
P_1	5
P_2	3
P_3	1

Gantt chart:



Ready queue:



∴ completion time of each process is

$P_1 = 9$

$P_2 = 8$

$P_3 = 5$

∴ P_3 will complete first ..

(3) Consider the following processes scheduled using FCFS. All processes arrive at time 0.

process	Burst time
P1	6 ms
P2	2 ms
P3	8 ms
P4	3 ms

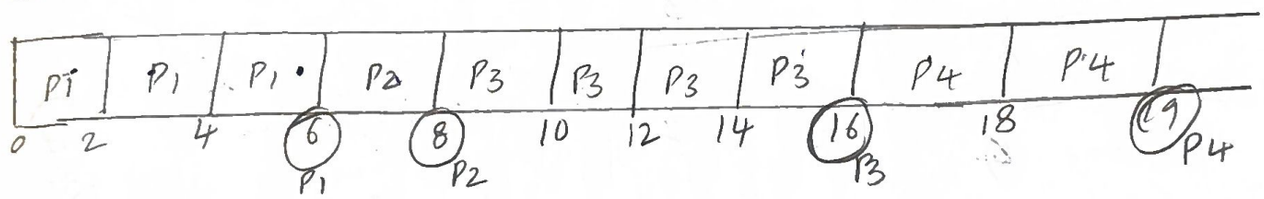
(a)

given

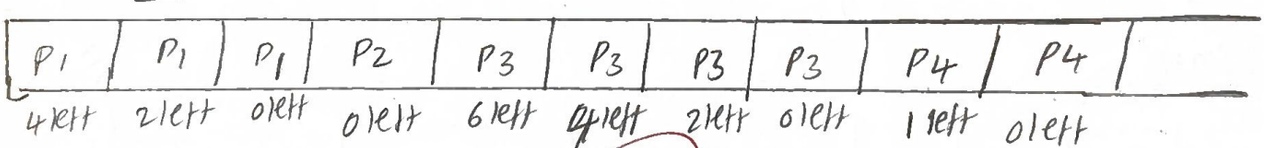
process	Burst time
P1	6 ms
P2	2 ms
P3	8 ms
P4	3 ms

time quantum = 2 ms

Gantt chart:



Ready queue



Process	Burst time	Arrival time	Completion time	TAT	Waiting time
P1	6	0	6	6	6-6 = 0
P2	2	0	8	8	8-2 = 6
P3	8	0	16	16	16-8 = 8
P4	3	0	19	19	19-3 = 16

Average waiting time = $\frac{0+6+8+16}{4} = \frac{30}{4} = 7.50$ ms.

Turnaround time = $\frac{6+8+16+19}{4} = \frac{49}{4} = 12.25$ ms.

④ Consider the following processes using FCFS scheduling

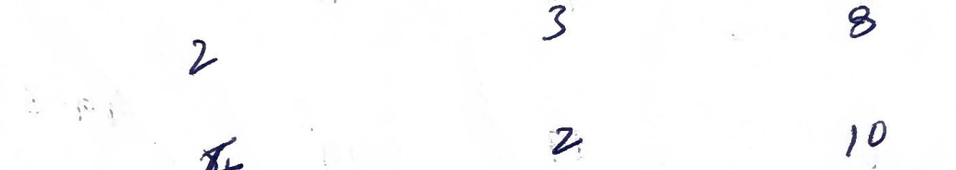
Process	Arrival time	Burst time
P_1	0	5
P_2	2	3
P_3	4	2

what is the average turnaround time.

Given.

process	Arrival time	Burst time
P_1	0	5
P_2	2	3
P_3	4	2

Gantt chart:



Ready queue

process	Arrival time	Burst time	completion	TAT
P_1	0	5	5	5-0=5
P_2	2	3	8	8-2=6
P_3	4	2	10	10-4=6

(completion - arrival time)

wait
5-5=0
6-3=3
6-2=4

∴ Average TAT = $\frac{5+6+6}{3} = \frac{17}{5} = 5.66$ ms.

average waiting time = $\frac{0+3+4}{3} = \frac{7}{3} \approx 2.33$ ms

(5) All processes arrive at time 0 and are scheduled using non-preemptive SJF. (Shortest Job first)

process	Burst time
P_1	7 ms
P_2	4 ms
P_3	1 ms
P_4	4 ms

Assuming ties are resolved by process order, what is the average waiting time?

Let; arrange the process in shorted order.

process	Burst time
P3	1 ms
P2	4
P4	4
P_1	7

Gantt chart:



Ready Queue:

Process	AT	Burst time	Completion	TAT	waiting
P_1	0	7	16	16	16 − 7 = 9
P_2	0	4	5	5	5 − 4 = 1
P3	0	1	1	1	1 − 1 = 0
P4	0	4	9	9	9 − 4 = 5

Average waiting time = $\frac{9+1+0+5}{4} = \frac{15}{4} = 3.75$ ms.

⑥ All processes arrive at time 0

Process	Burst time	priority
P_1	5 ms	3
P_2	3 ms	1
P_3	4 ms	2

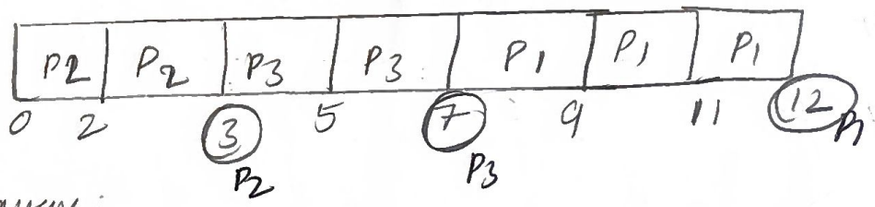
What is the average waiting time using non-preemptive priority scheduling.

let

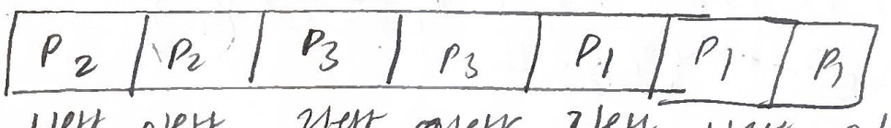
first arrange the processes by priority

process	Burst time
P_2	3
P_3	4
P_1	5

Gantt chart



Ready queue:



process	Burst time	Arrival	completion	TAT	Waiting
P_1	5	0	12	12	12-5=7
P_2	3	0	3	3	3-3=0
P_3	4	0	7	7	7-4=3

Average waiting time = $\frac{7+0+3}{3} = \frac{10}{3} = 3.333..$ ms.

(7). Consider three processes arriving at time 0 and scheduled using Round Robin with a time quantum of 2 ms.

process	Burst time
P_1	5 ms
P_2	3 ms
P_3	1 ms

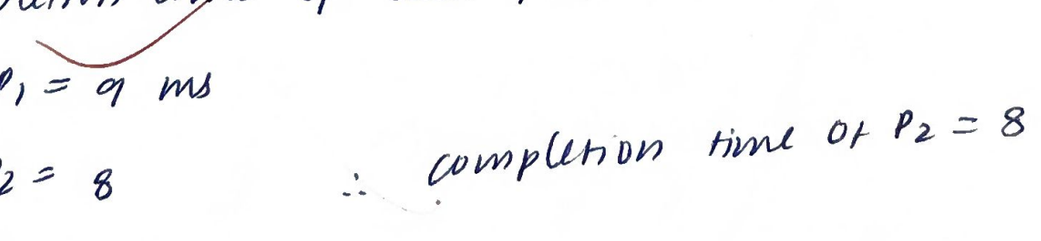
Assume the ready queue initially contains P_1, P_2, P_3 in that order, what is the completion time of P_2.

Given

process	Burst time
P_1	5 ms
P_2	3 ms
P_3	1 ms

Time quantum = 2 ms.

Gantt chart:



Ready queue:

∴ Completion time of each process is

$P_1 = 9$ ms

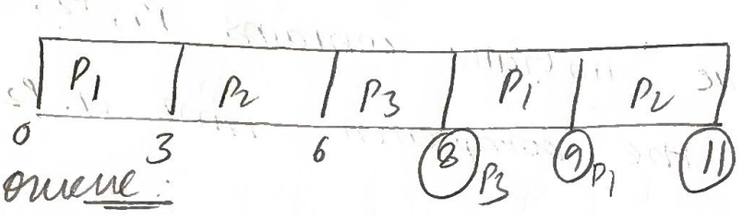
$P_2 = 8$ ∴ completion time of $P_2 = 8$ ms.

$P_3 = 5$

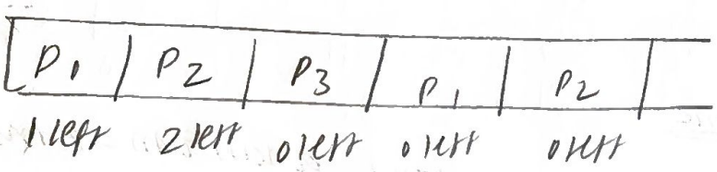
(8) Three process arrive at time 0 and are scheduled using Round Robin with a time quantum = 3 ms

Process	Burst time
P_1	4 ms
P_2	5 ms
P_3	2 ms.

Gantt chart:



Ready queue:



Process	Burst	Arrival	completion	TAT	WT
P_1	4	0	9	9	9 − 4 = 5
P_2	5	0	11	11	11 − 5 = 6
P_3	2	0	8	8	8 − 2 = 6

∴ completion time of P_3 = 8 ms

(8) three processes arrive at different times and are scheduled using round robin scheduling with a time quantum of 2ms.

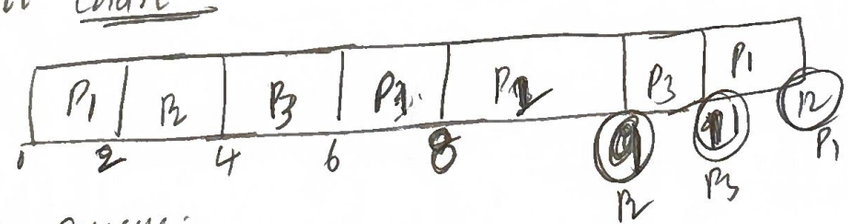
Process	Arrival time	Burst time
P_1	0	5
P_2	1	3
P_3	2	4

Draw the Gantt chart and determine the completion times of P2

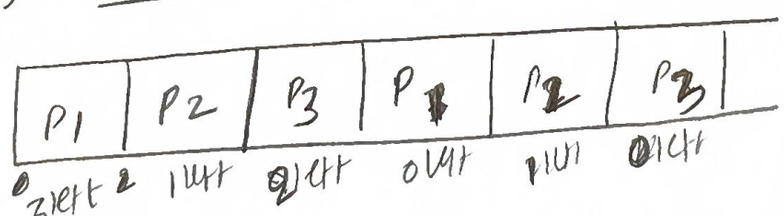
for

process	Arrival time	Burst time
P_1	0	5
P_2	1	3
P_3	2	4

Gantt chart:



ready queue:



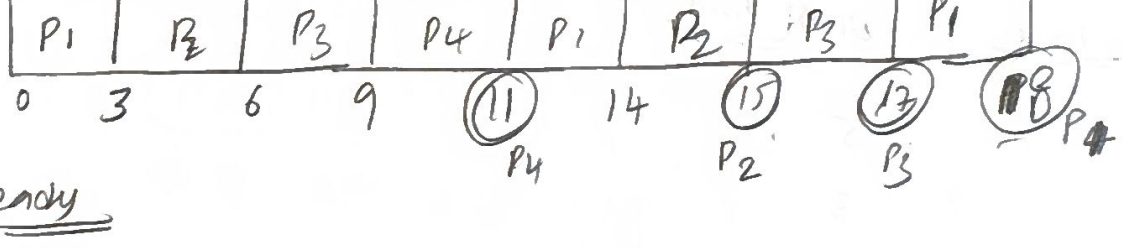
Process	Arrival time	Burst time	completion time	TAT	WT
P_1	0	5	12	12	11
P_2	1	3	9	8	8
P_3	2	4	11	9	9

(10). Four processes are scheduled using Round Robin scheduling with a time quantum of 3 ms.

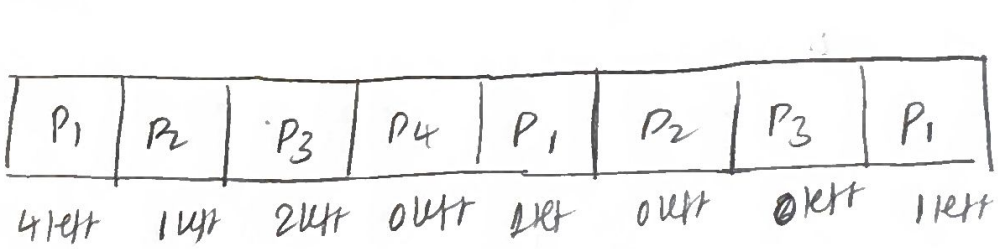
Process	Arrival Time	Burst Time
P_1	0	7
P_2	2	4
P_3	3	5
P_4	5	2

Draw the gantt chart and calculate the average waiting time of all processes.

Gantt:



Ready



Processes	Arrival time	Burst time	completion	TRT	waiting time
P_1	0	7	18	18	11
P_2	2	4	15	13	9
P_3	3	5	17	14	9
P_4	5	2	11	6	4

Average wt $= \frac{11+9+9+4}{4} = \frac{33}{4} = 8.25$

4)33(8.25
32
10

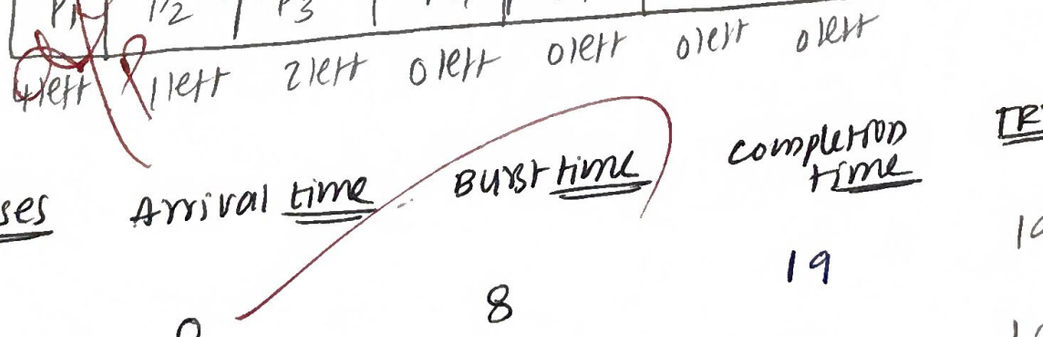
(11) Find the average turnaround time.

Four processes arrive at different times and are executed using Round Robin scheduling with a time quantum of 4 ms.

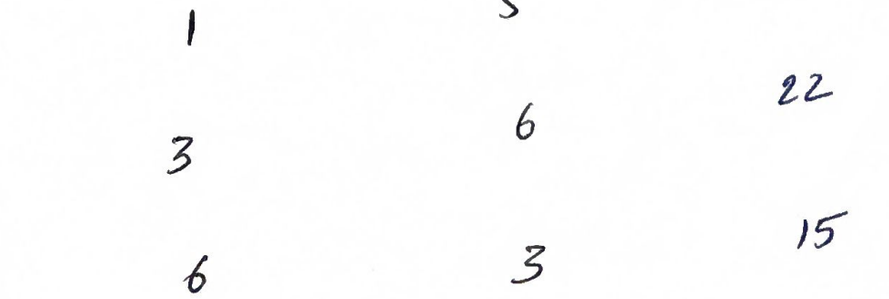
Process	Arrival time	Burst time
P_1	0	8
P_2	1	5
P_3	3	6
P_4	6	3

(a) construct the gantt chart and calculate average turnaround time.

Sol Gantt chart:



ready queue:



Processes	Arrival time	Burst time	completion time	TRT	WT
P_1	0	8	19	19	11
P_2	1	5	20	19	14
P_3	3	6	22	19	13
P_4	6	3	15	9	6